

01- Feature List

Feature List

1. The Doctor's Interface (The AI Co-Pilot)

- **Record consultation & create report (AI):** The doctor dictates or types rough notes, and the AI automatically formats it into a professional, structured medical report.
- **Summary & Decision Support (AI):** The RAG module reads the patient's history, summarizes it, and actively flags potential errors (e.g., *"Warning: You prescribed Medication A, but the patient's history shows an allergy"*).
- **Billing identification (AI):** The AI reads the generated report, identifies the correct billing codes, and presents them. The doctor just clicks "Approve."

2. The Patient's Interface (The Smart Portal)

- **Profile updates after consultation (AI):** *This is a genius feature.* If the doctor writes in the consultation notes, *"Patient mentioned she has a new phone number: 0152-XXXX,"* the AI automatically extracts that data and updates the patient's PostgreSQL profile. The patient just sees their updated profile without anyone manually typing it.
- **Report simplification (AI) & Q&A (AI):** The AI takes the complex report the doctor just generated and translates it into simple terms for the patient to read securely at home.
- **Scheduling & Check-in:** Voice AI handles the calls, and the mobile app handles the physical check-in.

3. The Receptionist / Clinic Staff (The AI Manager)

- **Time management & Work scheduling (AI):** Instead of a human manager figuring out who works on Tuesday, the AI looks at the staff availability, handles sick days, and calculates who should replace who.

- **Onboarding (AI):** A dedicated RAG chatbot just for the staff. A new nurse can ask, *"How do I process a blood sample?"* and the AI answers based on the clinic's internal rulebook.
- **Invoice Generation:** Pure logic. The system takes the doctor's approved AI codes and generates/emails the PDF.

Security & Auth: Feature 1

Patient Portal: Features 4, 5, 6, 13, 14, 15, 16

Receptionist/Admin: Features 3, 7, 8, 9, 18, 19

Doctor/AI Logic: Features 2, 10, 11, 12, 17

02- Priorities

Priority List (MoSCoW)

1. "M" - MUST HAVE (The Core Engine & Thesis Requirements)

- **Doctor:** Authentication, Document consultation & create report (AI).
- **Receptionist:** Authentication, Manual Check-in, patient registration, book/update/delete appointment, Invoice Generation (Pure Logic).
- **Patient:** Authentication, Self check-in (Pure Logic/Mobile), Web Appointment Scheduling (Pure Logic), update/cancel appointment

2. "S" - SHOULD HAVE (The High-Value AI Magic)

- **Doctor:** Summary of patient state , Decision Support (AI RAG), Billing identification (AI code suggestion).
- **Patient:** Appointment scheduling (Voice AI Call Agent), Report simplification (LLM) / explainability (AI RAG).

3. "C" - COULD HAVE (The "Bonus" Features if time permits)

- **Patient:** Profile updates automatically extracted from the consultation report (AI), Q&A chatbot / **to verify**
- **Doctor:** On-Demand Q&A Chatbot/ **to verify**
- **Receptionist:** Time management & Work scheduling (Pure Logic / or AI).

4. "W" - WON'T HAVE (Out of Scope for Version 1.0)

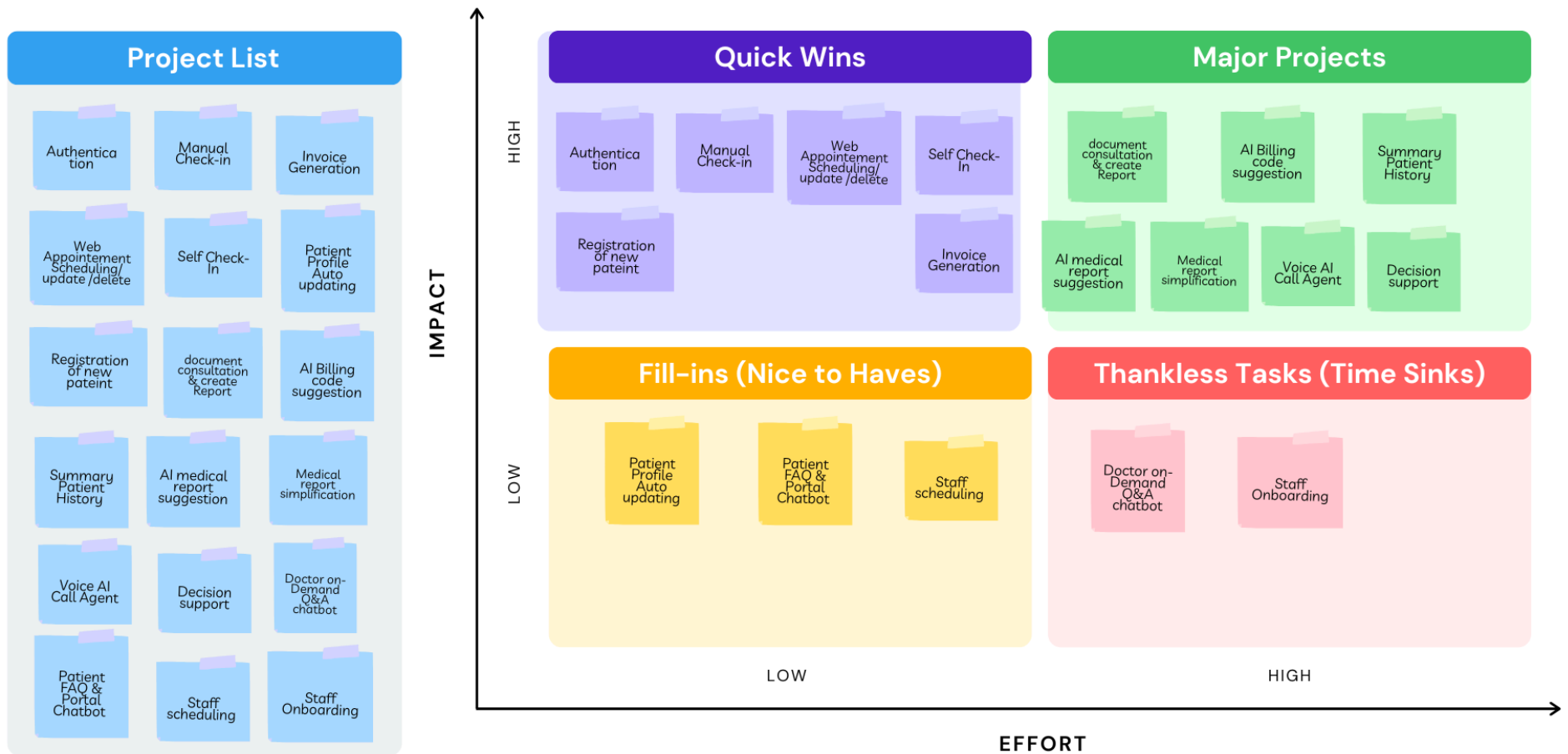
- **Receptionist:** Staff Onboarding AI Chatbot (We will save this for Version 2.0 because the patient-facing AI is more important right now).

Business Value is based on the MoSCoW Priority, and **Effort/Complexity** is estimated based on the type of technology required (Pure Logic vs. LLM Extraction/Generation vs. RAG/Optimization).

	Low Effort / Complexity (Pure Logic / CRUD)	Moderate Effort / Complexity (LLM/NLP Generation & Extraction)	High Effort / Complexity (RAG, Voice AI, Optimization)
HIGH VALUE (MUST HAVE)	* Authentication * Manual Check-in / Registration * Invoice Generation (Pure Logic) * Web Appointment Scheduling * Mobile Self Check-in	* Record consultation & create report (AI generation)	
MEDIUM VALUE (SHOULD HAVE)		* AI Billing Code Suggestion (NLP/LLM) * AI Medical Report Simplification (LLM)	* AI Patient History Summary & Decision Support (RAG) * Voice AI Call Agent
LOW VALUE (COULD HAVE / V2.0)		* AI-Driven Patient Profile Auto-Updating (LLM Extraction)	* Doctor On-Demand Q&A Chatbot (Conversational RAG) * Patient FAQ & Portal Chatbot (RAG) * AI-Assisted Staff Scheduling & Conflict Resolution * Staff Onboarding AI Chatbot (RAG) (Deferred to V2.0)

impact/effort Matrix

Impact/Effort Matrix



03- Use Cases

MUST HAVE

Feature 1

Feature 1 : Role-Based Authentication

1. Feature Description

The system must verify the identity of any user attempting to access the platform and ensure they can only view and interact with the interfaces authorized for their specific job or role (Doctor, Receptionist, or Patient).

2. Actors Involved

- Doctor
- Receptionist (Clinic Staff)
- Patient

3. Use Case Flow: UC-AUTH-01 (User Login)

- **Precondition:** The user is not logged in and is on the system's public landing page.
- **Main Success Scenario:**
 - The user navigates to the login screen.
 - The system prompts the user to enter their credentials (e.g., email/password or phone/OTP).
 - The user enters their credentials and clicks "Submit."
 - The system validates the credentials against its records.
 - The system identifies the specific role assigned to that user.
 - The system grants access and automatically redirects the user to their designated dashboard (e.g., Clinical Workspace for the Doctor, Admin Dashboard for the Receptionist).
- **Alternate Flow (Failure):**
 - If the credentials do not match, the system denies access, remains on the login screen, and displays an error message: *"Invalid credentials. Please try again."*
- **Postcondition:** The user is securely authenticated and active within their permitted workspace.

4. Formal Functional Requirements

- **REQ-AUTH-01:** The system shall provide a secure login interface for all authorized actors.

- **REQ-AUTH-02:** The system shall validate user credentials before granting access to any internal module.
- **REQ-AUTH-03:** The system shall enforce Role-Based Access Control (RBAC) to ensure users can only access features permitted by their assigned role.
- **REQ-AUTH-04:** The system shall automatically redirect successfully authenticated users to their role-specific landing page.

Feature 2

Feature 2 : Document Consultation & Report Generation

1. Feature Description

The system must allow the Doctor to manually type raw, unstructured consultation notes. The system will securely process this text through an AI module to automatically generate a formal, structured medical report. The Doctor retains final authority to review, edit, and approve the report before it is saved to the database.

2. Actors Involved

- Doctor

3. Use Case Flow: UC-DOC-01 (Generate Medical Report)

- **Precondition:** The Doctor is authenticated and in an active consultation session.
- **Main Success Scenario:**
 1. The Doctor types rough notes into the "Consultation Notes" text area.
 2. The Doctor clicks "Generate Formal Report."
 3. The system securely sends the text to the AI module.
 4. The AI returns a professionally formatted medical report.
 5. The system displays the report in an editable text box.
 6. The Doctor reviews, makes any manual corrections, and clicks "Approve & Save."
 7. The system saves the final document to the Patient's medical record.
- **Alternate Flow 1 (Save Without AI):**
 - The Doctor types notes into the text area but decides a formal AI report is unnecessary.
 - The Doctor clicks "Save Raw Notes."
 - The system bypasses the AI module and saves the unstructured text directly to the Patient's medical record.
- **Alternate Flow 2 (Auto-Save Draft):**
 - As the Doctor types into the "Consultation Notes" box, the system automatically saves the text as a "Draft" every 30 seconds. If the

session is accidentally closed, the Doctor can recover the draft upon reopening the patient's file.

4. Formal Functional Requirements

- **REQ-DOC-01:** The system shall provide a text input interface for the Doctor to record unstructured notes.
- **REQ-DOC-02:** The system shall utilize an AI module to translate unstructured notes into a structured medical report.
- **REQ-DOC-03:** The system shall display the AI-generated report in an editable format prior to saving.
- **REQ-DOC-04:** The system shall require the explicit manual approval of the Doctor before finalizing the report.

Feature 3

Feature 3: Invoice Generation & Billing Management (Public , Private Insurance or Self-Payers)

1. Feature Description

The system must allow the Receptionist to finalize the billing process for a completed medical appointment based on the patient's insurance type. For Private/Self-Pay patients, the system generates a formal, legally compliant PDF invoice. For Public (GKV) patients, the system validates the EBM codes and queues the data for the quarterly batch submission (*Quartalsabrechnung*).

2. Actors Involved

- Receptionist (Clinic Staff)

3. Use Case Flow: UC-REC-01 (Generate Patient Invoice)

- **Precondition:** *The Doctor has marked an appointment as "Completed" with approved billing codes attached (Feature 12).*
- **Main Success Scenario:**
 1. *The Receptionist navigates to the "Billing Dashboard."*
 2. *The system displays a list of completed appointments awaiting processing.*
 3. *The Receptionist selects a specific Patient's completed appointment.*
 4. *The system automatically detects the Patient's Insurance Type:*
 - **Path A (Private / GOÄ):**
 1. *The system calculates the total fee.*
 2. *The system generates a formatted PDF invoice.*
 3. *The Receptionist clicks "Send/Print PDF."*
 4. *The appointment status updates to "Pending Patient Payment."*
 - **Path B (Public / EBM):**
 1. *The system validates the EBM codes (no total monetary fee is shown to the patient).*
 2. *The system queues the record into the "KV Batch Pool" for end-of-quarter submission.*
 3. *The Receptionist clicks "Approve for KV."*

4. The appointment status updates to "Ready for KV Billing." (No PDF is generated).

- **Alternate Flow (Missing Codes):**

- If no billing codes are attached, the system blocks processing and displays: "Cannot process billing. Please request codes from the treating Doctor."

- **Postcondition:**

- **For Private / Self-Payers (PKV):** A legally compliant PDF invoice is generated, permanently logged in the clinic's financial database, and the appointment's billing status is updated to "Pending Payment".
- **For Public Insurance (GKV):** No PDF is generated. The patient's billing record is validated, locked, and appended to the quarterly batch export queue (Quartalsabrechnung). The appointment's billing status is updated to "Ready for KV Billing".(the government)

4. Formal Functional Requirements

- **REQ-REC-01:** The system shall display a unified list of completed appointments awaiting billing finalization.
- **REQ-REC-02:** The system shall dynamically route the billing workflow based on the patient's insurance status (Private vs. Public).
- **REQ-REC-03:** For Private/Self-Pay patients, the system shall calculate total costs and generate a downloadable PDF invoice containing payment instructions.
- **REQ-REC-04:** For Public (GKV) patients, the system shall suppress PDF generation and instead flag the data record as ready for batch export (Quartalsabrechnung).
- **REQ-REC-05:** The system shall prevent billing finalization if no codes are associated with the appointment.

Feature 4

Feature 4: Web Appointment Scheduling

1. Feature Description

The system must provide a web-based, self-service portal allowing authenticated Patients to view a real-time calendar of available consultation slots. The Patient can select an open time slot, confirm the booking, and the system will automatically update the clinic's central database to secure the appointment.

2. Actors Involved

- Patient

3. Use Case Flow: UC-PAT-02 (Book Web Appointment)

- **Precondition:** The Patient is authenticated and has accessed the "New Appointment" section of the Patient Portal.
- **Main Success Scenario:**
 - The Patient selects the desired Doctor (if it is a multi-doctor clinic) or appointment type.
 - The system queries the database and displays a calendar with available dates and time slots for the upcoming weeks.
 - The Patient clicks on an available time slot (e.g., "Tuesday, 14:00").
 - The system prompts the Patient to briefly type the reason for the visit and click "Confirm Booking."
 - The system performs a final real-time check to ensure the slot is still available.
 - The system locks the time slot, creates a new appointment record with the status **Scheduled**, and links it to the Patient's ID and Doctor's ID.
 - The system displays a success confirmation and updates the Patient's dashboard.
- **Alternate Flow (The "Race Condition" / Double Booking):**
 - Between Step 3 and Step 4, another patient books the exact same 14:00 time slot.
 - At Step 5, the system detects the slot is no longer available.

- The system blocks the booking, does *not* create the appointment, and alerts the user: *"We're sorry, this time slot was just booked by someone else. Please select another time."* The calendar refreshes.

4. Formal Functional Requirements

- **REQ-SCHED-01:** The system shall display a real-time calendar of available appointment slots for a selected doctor or date range.
- **REQ-SCHED-02:** The system shall allow the Patient to input a brief text description for the reason of their visit.
- **REQ-SCHED-03:** The system shall implement concurrency controls to prevent two users from booking the same time slot simultaneously (double-booking).
- **REQ-SCHED-04:** The system shall create a new appointment record in the database with a **Scheduled** status upon successful confirmation.
- **REQ-SCHED-05:** The system shall immediately remove the booked slot from the public availability calendar.

Feature 5

Feature 5: Patient Mobile Self Check-in

1. Feature Description

The system must allow scheduled Patients to check themselves in upon arriving at the clinic using their own mobile device (e.g., by scanning a QR code in the waiting room or clicking a secure link sent via SMS). The system will validate that the appointment is for today, update the internal database status, and immediately notify the clinic staff that the patient has arrived, bypassing the reception desk entirely.

2. Actors Involved

- Patient

3. Use Case Flow: UC-PAT-01 (Mobile Appointment Check-in)

- **Precondition:** The Patient is authenticated (via phone OTP) and has a valid appointment scheduled for the current day.
- **Main Success Scenario:**
 - The Patient arrives at the clinic and scans the check-in QR code at the entrance with their smartphone.
 - The system opens the Patient Portal and prompts the user to authenticate (if not already logged in).
 - The system displays the Patient's appointment details for today (e.g., "Dr. Smith - 14:00").
 - The Patient clicks the "Confirm Arrival / Check-in" button.
 - The system validates the current time against the appointment time.
 - The system updates the appointment status in the database from **Scheduled** to **Waiting**.
 - The system displays a success message: *"You are checked in. Please take a seat."*
 - The Receptionist and Doctor dashboards automatically update to show the Patient is in the waiting room.
- **Alternate Flow (Too Early / Wrong Day):**
 - If the Patient tries to check in for an appointment that is scheduled for tomorrow, the system blocks the action and displays: *"Check-in is only available on the day of your appointment."*

- **Alternate Flow (Already Checked In):**
 - If the Patient clicks the button twice, the system recognizes the state is already **Waiting** and simply displays: *"You are already checked in."*

4. Formal Functional Requirements

- **REQ-PAT-01:** The system shall provide a mobile-responsive interface for patients to view their upcoming appointments.
- **REQ-PAT-02:** The system shall restrict the check-in action to a specific time window (e.g., only on the date of the appointment).
- **REQ-PAT-03:** The system shall update the database status of the appointment from **Scheduled** to **Waiting** upon successful patient confirmation.
- **REQ-PAT-04:** The system shall reflect this status change in real-time on the authorized Receptionist and Doctor dashboards.
- **REQ-PAT-05:** The system shall prevent duplicate check-ins for the same appointment.

Feature 6

Feature 6: Appointment Modification & Cancellation

1. Feature Description

The system must allow authenticated Patients to view their upcoming scheduled appointments and provide the ability to cancel or reschedule them through the web portal. Upon cancellation, the system must immediately free up that specific time slot in the doctor's calendar so other patients can book it.

2. Actors Involved

- Patient

3. Use Case Flow: UC-PAT-03 (Cancel/Reschedule Appointment)

- **Precondition:** The Patient is authenticated and has at least one upcoming appointment with the status **Scheduled**.
- **Main Success Scenario (Cancellation):**
 - The Patient navigates to the "My Appointments" dashboard.
 - The Patient clicks the "Cancel Appointment" button next to an upcoming visit.
 - The system displays a confirmation prompt: *"Are you sure you want to cancel this appointment?"*
 - The Patient clicks "Confirm."
 - The system updates the appointment database status from **Scheduled** to **Cancelled**.
 - The system immediately releases the associated time slot back into the public availability calendar.
 - The Patient's dashboard refreshes to show the appointment is cancelled.
- **Alternate Flow (Reschedule):**
 - If the Patient clicks "Reschedule," the system holds the current appointment, opens the booking calendar (UC-PAT-02), allows them to pick a new time, and then automatically cancels the old slot once the new one is confirmed.
- **Alternate Flow (Too Late to Cancel):**

- *Business Rule:* The system checks the current time. If the appointment is less than 24 hours away, the system disables the cancel button and displays: *"Appointments within 24 hours must be cancelled by calling the clinic directly."*

4. Formal Functional Requirements

- **REQ-MOD-01:** The system shall display a list of all upcoming appointments for the authenticated Patient.
- **REQ-MOD-02:** The system shall provide a UI control for the Patient to cancel a **Scheduled** appointment.
- **REQ-MOD-03:** The system shall update the database record to **Cancelled** and remove it from the Doctor's active daily schedule.
- **REQ-MOD-04:** The system shall instantly re-expose the cancelled time slot as "Available" in the scheduling database.
- **REQ-MOD-05:** The system shall prevent self-service cancellations within a predefined time window (e.g., 24 hours prior to the appointment).

Feature 7

Feature 7: Receptionist Manual Check-in

1. Feature Description

The system must provide the Receptionist with the ability to manually update a Patient's appointment status upon their physical arrival at the clinic. This serves as the primary fallback for patients who are unable or unwilling to use the mobile self check-in feature.

2. Actors Involved

- Receptionist (Clinic Staff)

3. Use Case Flow: UC-REC-02 (Manual Patient Check-in)

- **Precondition:** The Receptionist is authenticated, and the Patient has a valid **Scheduled** appointment for today.
- **Main Success Scenario:**
 - The Patient physically arrives at the reception desk and provides their name.
 - The Receptionist navigates to the "Daily Schedule" dashboard.
 - The system displays a chronological list of all appointments for the current day.
 - The Receptionist locates the specific Patient's appointment.
 - The Receptionist clicks the "Mark as Arrived" (or "Check-in") button next to the patient's name.
 - The system updates the appointment database status from **Scheduled** to **Waiting**.
 - The system visually updates the Doctor's dashboard to indicate the patient is ready to be seen.
- **Alternate Flow (Patient Arrives on the Wrong Day):**
 - If the Patient arrives, but the Receptionist sees their appointment is actually scheduled for tomorrow, the Receptionist can use the system to either reschedule them for today (if there is a free slot) or inform the patient of the error.

4. Formal Functional Requirements

- **REQ-REC-06:** The system shall display a daily schedule view for the Receptionist, listing all patient appointments for the current date.
- **REQ-REC-07:** The system shall provide a manual UI control (e.g., a button) for the Receptionist to change an appointment's status to **Waiting**.
- **REQ-REC-08:** The system shall synchronize this status change in real-time with the Doctor's clinical dashboard.

Feature 8

Feature 8: Receptionist Patient Registration

1. Feature Description

The system must provide the Receptionist with an administrative interface to manually create a new Patient profile in the database. The system must be inclusive of all patient demographics, supporting both "digital" patients (who receive portal access via email/SMS) and "analog" patients (who only have a physical address and landline, requiring manual clinic management).

2. Actors Involved

- Receptionist (Clinic Staff)

3. Use Case Flow: UC-REC-03 (Register New Patient)

- **Precondition:** The Receptionist is authenticated and a new patient is either on the phone or standing at the front desk.
- **Main Success Scenario:**
 - The Receptionist navigates to the "Patient Directory" dashboard and clicks "Add New Patient."
 - The system displays a blank patient registration form.
 - The Receptionist inputs the patient's mandatory demographic data (First Name, Last Name, Date of Birth, Physical Address).
 - The Receptionist inputs optional digital contact data (Email, Mobile Phone) if the patient has them, along with Private Insurance details.
 - The Receptionist clicks "Save Profile."
 - The system validates that all mandatory fields are filled.
 - The system creates a new record in the **Patients** database table and assigns a unique **Patient_ID**.
 - **System Check:** If an Email or Mobile Phone was provided, the system automatically generates a temporary password/OTP and sends it to the patient. If *no* digital contact was provided, the system skips this step.
 - The Receptionist's screen redirects to the newly created Patient Profile.

- **Alternate Flow (Duplicate Patient):**
 - Between Step 5 and 6, the system checks the database for matching names and Dates of Birth.
 - If a match is found, the system pauses and displays a warning: *"A patient with the name Thomas Müller born on 14.05.1980 already exists. Do you want to open their existing file or create a new one?"*

4. Formal Functional Requirements

- **REQ-REC-09:** The system shall provide a data entry form for the Receptionist, enforcing mandatory fields (Name, DOB, Physical Address) while leaving digital contact fields (Email, Mobile) as optional.
- **REQ-REC-10:** The system shall validate all required fields before attempting to save the record.
- **REQ-REC-11:** The system shall check for potential duplicate patient records based on Name and Date of Birth prior to creation.
- **REQ-REC-12:** The system shall generate a unique **Patient_ID** and store the new profile in the centralized database.
- **REQ-REC-13:** **IF** a valid email or mobile number is provided, the system shall automatically provision portal access credentials. **IF NOT**, the system shall store the patient record purely for internal clinic use without generating portal access.

Feature 9

Feature 9: Receptionist Appointment Management

1. Feature Description

The system must provide the Receptionist with administrative privileges to manually modify, reschedule, or cancel any existing patient appointment. This ensures the clinic maintains total control over its calendar when patients call in via phone or when the Doctor's schedule unexpectedly changes.

2. Actors Involved

- Receptionist (Clinic Staff)
- (Secondary) Patient - receives notification of changes.

3. Use Case Flow: UC-REC-04 (Modify or Cancel Appointment)

- **Precondition:** The Receptionist is authenticated and viewing the clinic's master calendar or a specific Patient's file.
- **Main Success Scenario (Cancellation via Phone):**
 - A patient calls the clinic to cancel their appointment.
 - The Receptionist searches for the Patient's name or navigates to the specific date on the master calendar.
 - The Receptionist selects the **Scheduled** appointment and clicks "Cancel Appointment."
 - The system updates the appointment status in the database from **Scheduled** to **Cancelled**.
 - The system immediately releases the associated time slot back into the public availability calendar.
 - The system automatically triggers an email/SMS confirmation to the Patient: *"Your appointment on [Date] has been successfully cancelled."*
- **Alternate Flow (Clinic-Initiated Reschedule):**
 - If the Doctor calls in sick, the Receptionist selects all appointments for that day and clicks "Reschedule."
 - The Receptionist selects new dates/times for the affected patients while on the phone with them.

- The system updates the existing appointment records with the new time slots and sends updated confirmation messages to the patients.

4. Formal Functional Requirements

- **REQ-REC-14:** The system shall allow the Receptionist to override, modify, or cancel any patient appointment regardless of how it was originally booked.
- **REQ-REC-15:** The system shall immediately free up the associated calendar time slot when an appointment is cancelled by the Receptionist.
- **REQ-REC-16:** The system shall maintain an audit trail (log) indicating that the cancellation/modification was performed by a staff member, not the patient.
- **REQ-REC-17:** The system shall automatically send a notification (Email/SMS) to the patient whenever their appointment is modified or cancelled by clinic staff.

SHOULD HAVE

Feature 10

Feature 10: AI Patient History Summary

1. Feature Description

The system must utilize a Retrieval-Augmented Generation (**RAG**) AI module to automatically synthesize a patient's past medical history. When a Doctor opens a returning patient's file, the system retrieves past consultation reports and extracts key data (chronic conditions, past surgeries, active medications, and allergies) into a concise, instantly readable "Patient Briefing" dashboard panel, saving the Doctor from reading through old documents manually.

2. Actors Involved

- Doctor
- AI Module (**RAG**)

3. Use Case Flow: UC-DOC-03 (Auto-Generate Patient Summary)

- **Precondition:** The Doctor is authenticated, and the patient has at least one past medical report saved in the database (from Feature 2).
- **Main Success Scenario:**
 - The Doctor clicks on the Patient's name from the daily schedule to open their clinical file.
 - The system loads the UI and simultaneously triggers a background request to the RAG module.
 - The vector database retrieves the patient's most clinically relevant past consultation notes and reports.
 - The LLM synthesizes this unstructured text into a structured JSON format (e.g., `{"allergies": ["Penicillin"], "chronic_conditions": ["Asthma"], "recent_visits": "Follow-up for sinus infection on Feb 12"}`).

- The system renders this data in a clean "Patient Briefing" UI panel at the top of the screen.
- The Doctor reads the summary in seconds before beginning the new consultation.
- **Alternate Flow (New Patient / Empty State):**
 - If the system detects zero past records in the database for this **Patient_ID**, it skips the AI generation entirely to save processing costs.
 - The UI panel simply displays: *"New Patient. No historical data available."*
- **Alternate Flow (AI Processing Delay):**
 - If the LLM takes more than 2 seconds to generate the summary, the UI displays a subtle "Summarizing past records..." loading skeleton so the Doctor knows the system is working, rather than freezing the screen.

4. Formal Functional Requirements

- **REQ-SUM-01:** The system shall automatically trigger the summary generation process upon the opening of a patient's clinical file without requiring a manual click.
- **REQ-SUM-02:** The RAG module shall strictly limit its retrieval scope to the specific **Patient_ID** to prevent cross-contamination of medical records.
- **REQ-SUM-03:** The system shall structure the AI output into distinct, readable categories (e.g., Allergies, Chronic Conditions, Recent History).
- **REQ-SUM-04:** The system shall handle "empty states" gracefully for new patients, bypassing the LLM call entirely.
- **REQ-SUM-05:** The system shall permanently display a liability disclaimer below the summary stating: *"AI-generated summary. Always verify critical details with the patient."*

Feature 11

Feature 11: AI Proactive Decision Support

1. Feature Description

The system must provide real-time, context-aware clinical decision support during the consultation. As the Doctor types consultation notes or treatment plans, the AI module continuously analyzes the text in the background. It cross-references the current input against the patient's known medical history (Feature 10) and standard medical guidelines. If a critical contraindication is detected (e.g., drug-allergy interaction), the system displays a highly visible, non-intrusive warning to the Doctor.

2. Actors Involved

- Doctor
- AI Module (**RAG** / Semantic Analysis)

3. Use Case Flow: UC-DOC-04 (Receive Proactive Clinical Alerts)

- **Precondition:** The Doctor is authenticated, actively typing in the "Consultation Notes" interface (Feature 2), and the patient's historical data is loaded into the session.
- **Main Success Scenario:**
 - The Doctor types a proposed treatment into the notes (e.g., *"Prescribing 500mg Amoxicillin"*).
 - The system captures the text input and sends it to the AI module in the background.
 - The AI identifies the medical entity ("Amoxicillin") and cross-references it against the patient's profile (which lists "Allergy: Penicillin").
 - The AI detects a critical conflict.
 - The system immediately renders a highly visible warning banner near the text box: "**⚠️ ALERT: Patient has a documented allergy to Penicillin-class antibiotics (Amoxicillin).**"
 - The Doctor sees the alert, deletes "Amoxicillin," and types a safe alternative. The warning disappears.
- **Alternate Flow (No Conflict / Silent Operation):**

- If the Doctor types a safe treatment, the AI validates it but the UI does absolutely nothing. The system remains invisible so it does not interrupt the Doctor's workflow.
- **Alternate Flow (Doctor Override):**
 - The Doctor sees the warning but uses their professional judgment to proceed anyway (e.g., the benefit outweighs the risk).
 - The Doctor clicks "Dismiss Alert."
 - The system logs the dismissal in the backend audit trail (for liability protection) and hides the warning.

4. Formal Functional Requirements

- **REQ-DS-01:** The system shall analyze the Doctor's unstructured text input in near real-time to identify medical entities (medications, diagnoses, treatments).
- **REQ-DS-02:** The system shall cross-reference identified entities against the patient's documented allergies and chronic conditions to detect contraindications.
- **REQ-DS-03:** The UI shall display warnings in a non-blocking manner, allowing the Doctor to continue typing without forcing them to click a popup modal.
- **REQ-DS-04:** The system shall allow the Doctor to manually dismiss an active alert.
- **REQ-DS-05:** The system shall silently log all triggered alerts and manual dismissals into a secure database audit trail.

Feature 12

Feature 12: AI Billing Code Suggestion

1. Feature Description

Upon the completion of a patient consultation, the system utilizes an AI module (NLP/LLM) to analyze the finalized medical report. The system first checks the patient's insurance status (**Public/GKV or Private/PKV**). The AI then identifies the specific examinations, treatments, and procedures performed, and maps them to the appropriate standard billing catalog (**EBM for public, GOÄ for private**). The system presents these suggestions to the Doctor for mandatory review, editing, and final approval before the data is passed to the Receptionist for final invoice generation.

2. Actors Involved

- Doctor
- AI Module (Information Extraction / Classification)
- **Patient Database (to verify insurance type)**

3. Use Case Flow: UC-DOC-06 (Review and Approve AI Billing Suggestions)

1. **Precondition:** The Doctor has just clicked "Approve & Save" on the AI-generated medical report (completing Feature 2).
2. **Main Success Scenario:**
 - The system automatically routes the finalized report text to the AI Billing module.
 - The system queries the database to determine if the patient is GKV (Public) or PKV/Selbstzahler (Private).
 - The AI scans the text for actionable procedures (e.g., "Performed 15-minute detailed consultation," "Administered resting ECG").
 - The AI maps these actions to the correct catalog:
 - **If Private:** Generates GOÄ codes (e.g., GOÄ 3, GOÄ 651).
 - **If Public:** Generates EBM codes (e.g., EBM 03230, EBM 03320).

- The system displays this list in a clean UI panel titled "Suggested Billing Codes," indicating which catalog is being used.
- The Doctor reviews the list.
- The Doctor manually adds any missing codes or clicks an "X" to remove incorrect AI suggestions.
- The Doctor clicks "Approve for Invoicing."
- The system updates the consultation status to *Ready for Invoicing* and pushes the data to the Receptionist's queue.

3. Alternate Flow (AI Finds Nothing):

- If the consultation was so brief that the AI cannot confidently match any codes, the UI displays an empty list: *"No automatic codes detected. Please add codes manually."* The Doctor uses a search bar to add the codes themselves.

4. Formal Functional Requirements

- **REQ-BIL-01:** The system shall automatically parse the approved consultation report to extract billable medical procedures using an AI module.
- **REQ-BIL-02:** The system shall dynamically select the correct billing catalog (e.g., EBM for statutory insurance, GOÄ for private insurance) based on the patient's registered insurance profile.
- **REQ-BIL-03:** The system shall present the suggested codes in an interactive UI, allowing the Doctor to manually add, delete, or modify the selections.
- **REQ-BIL-04:** The system shall strictly enforce a "Human-in-the-Loop" architecture, requiring explicit manual approval from the Doctor before any billing data is finalized.
- **REQ-BIL-05:** Upon Doctor approval, the system shall seamlessly pass the authorized billing data to the Receptionist's Invoice Generation module.

Feature 13

Feature 13: Voice AI Call Agent

1. Feature Description

The system must integrate a conversational Voice AI Agent connected to the clinic's phone line. The Agent will use natural language processing (Speech-to-Text, LLM dialogue management, and Text-to-Speech) to understand patient requests, query the clinic's real-time scheduling database, and securely book appointments. The Agent will operate 24/7 and seamlessly transfer complex or unhandled calls to human staff during business hours.

2. Actors Involved

- Patient (via telephone)
- Voice AI Agent
- (Secondary) Receptionist (for human fallback)

3. Use Case Flow: UC-PAT-04 (Book Appointment via Voice AI)

- **Precondition:** The Patient dials the clinic's main or dedicated AI phone number.
- **Main Success Scenario:**
 - The Voice AI answers the call: *"Welcome to Clinic. I am the AI booking assistant. How can I help you today?"*
 - The Patient states their intent in natural language (e.g., *"I need to see Dr. Smith next Tuesday afternoon."*).
 - The AI extracts the intent (Booking), Doctor (Smith), and Timeframe (Tuesday afternoon).
 - The AI queries the centralized scheduling database (the exact same API built in Feature 4) to find available slots.
 - The AI verbally proposes an option: *"Dr. Smith has an opening next Tuesday at 14:00. Does that work for you?"*
 - The Patient confirms verbally (*"Yes, that's perfect."*).
 - The AI asks for the patient's name and date of birth to match/create their profile.

- The system locks the time slot, creates the appointment with the status **Scheduled**, and triggers an automated SMS confirmation to the Patient's phone.
- The AI politely ends the call.
- **Alternate Flow (No Availability):**
 - If Tuesday afternoon is booked, the AI proactively suggests the next closest availability: *"I'm sorry, Tuesday afternoon is fully booked. I have Wednesday morning at 09:00. Would you like that instead?"*
- **Alternate Flow (The "Human Handoff" / Fallback):**
 - *Business Rule:* If the Patient asks a medical question (*"My chest hurts, what should I do?"*), or if the AI fails to understand the patient after two attempts, the AI immediately aborts the booking flow.
 - The AI states: *"Let me connect you with our front desk staff who can help you further."*
 - The system routes the phone call directly to the Receptionist's line.

4. Formal Functional Requirements

- **REQ-VOI-01:** The system shall process inbound telephone calls using Speech-to-Text (STT) and Text-to-Speech (TTS) technologies.
- **REQ-VOI-02:** The system shall use an LLM to extract booking parameters (doctor, date, time) from natural conversational dialogue.
- **REQ-VOI-03:** The AI Agent shall interact with the core scheduling database in real-time to check availability and prevent double-booking.
- **REQ-VOI-04:** The system shall create a database record with a **Scheduled** status upon verbal confirmation from the caller.
- **REQ-VOI-05:** The system shall implement an automated fallback mechanism to route the call to a human Receptionist if the AI detects an emergency, a non-scheduling intent, or continuous confusion.

Feature 14

Feature 14: AI Medical Report Simplification

1. Feature Description

The system must provide an on-demand AI translation tool within the Patient Web Portal. When a patient views a finalized medical report from a past consultation, they can request the AI to simplify the clinical jargon into plain, non-technical language (e.g., 8th-grade reading level). The system will display this simplified explanation alongside the original legal document, empowering the patient to understand their health data without altering the official medical record.

2. Actors Involved

- Patient
- AI Module (LLM / Text Simplification)

3. Use Case Flow: UC-PAT-05 (Generate Simple Report Explanation)

- **Precondition:** The Patient is authenticated in the Web Portal and is viewing a finalized medical report (approved by the Doctor in Feature 2).
- **Main Success Scenario:**
 - The Patient navigates to the "My Health Records" dashboard and opens a specific consultation report.
 - The UI displays the original, technical report text.
 - The Patient clicks a button labeled "Explain in Simple Terms."
 - The system sends the original report text to the LLM with a strict system prompt (e.g., *"Translate this medical text for a layperson. Do not add any new medical advice. Maintain a neutral tone."*).
 - The AI returns the simplified text.
 - The system displays the simplified text in a distinct UI panel (e.g., a side-by-side view or a highlighted box) clearly separated from the original text.
 - The Patient reads the explanation to understand their diagnosis.
- **Alternate Flow (AI Service Unavailable):**
 - If the external LLM API times out or fails, the system catches the error gracefully and displays: *"Explanation service is temporarily*

unavailable. Please refer to the original document or contact the clinic if you have questions."

- **Alternate Flow (The Liability Guardrail):**
 - The system permanently affixes a warning banner to the simplified text: *"This is an AI-generated summary to help you understand your report. It is not medical advice. The original document remains the official clinical record."*

4. Formal Functional Requirements

- **REQ-SIMP-01:** The system shall integrate an LLM to perform on-demand text simplification of medical terminology into plain language.
- **REQ-SIMP-02:** The system shall execute the simplification process only upon explicit user request (on-demand) to optimize API usage and cloud costs.
- **REQ-SIMP-03:** The UI shall visually distinguish the AI-generated explanation from the original clinical report to prevent confusion regarding the official medical record.
- **REQ-SIMP-04:** The system prompt directing the LLM shall strictly prohibit the generation of novel medical advice, diagnosis, or treatment recommendations not present in the source text.
- **REQ-SIMP-05:** The system shall display a mandatory, un-hideable medical liability disclaimer alongside the simplified text.

COULD HAVE

Feature 15

Feature 15: AI-Driven Patient Profile Auto-Updating

1. Feature Description

The system utilizes an AI Information Extraction module to continuously keep the Patient's master profile up to date. When a new consultation report is finalized, the AI scans the unstructured text for changes in demographics (e.g., new address, phone number) or clinical baseline data (e.g., new allergies, changes in weight/height, smoking status). The system extracts these data points and proposes them as updates to the patient's master record, maintaining a strict audit trail of all automated changes.

2. Actors Involved

- Secondary System Actor: AI Module (Information Extraction / **LLM**)
- (Secondary) Patient / Doctor (for review)

3. Use Case Flow: UC-SYS-01 (Extract and Apply Profile Updates)

- **Precondition:** A Doctor has just finalized and saved a new medical report for the Patient (Feature 2).
- **Main Success Scenario:**
 - The system automatically routes the finalized report text to the AI extraction module in the background.
 - The AI scans the text specifically looking for entities related to the Patient's master profile (Contact Info, Vitals, Lifestyle, Allergies).
 - The AI identifies a new piece of data (e.g., text says *"Patient reports moving to Munich"*).
 - The AI formats this into a structured database update request:

```
{"field": "city", "old_value": "Karlsruhe",  
  "new_value": "Munich"}.
```
 - The system flags this update in the Patient's Web Portal under a "Suggested Profile Updates" notification.
 - The Patient logs in, sees the notification: *"Your doctor noted a change in your address. Update profile?"*
 - The Patient clicks "Confirm," and the database is officially updated.

- **Alternate Flow (Clinical Data - Doctor Approval):**
 - If the extracted data is *clinical* (e.g., a new allergy to Aspirin), the system does not ask the patient. It automatically adds it to the master clinical profile but tags it as **AI-Extracted**. It notifies the Doctor to verify it during the next visit.

4. Formal Functional Requirements

- **REQ-PROF-01:** The system shall utilize an NLP/LLM module to extract structured profile entities from unstructured consultation reports.
- **REQ-PROF-02:** The system shall categorize extracted data into Administrative (e.g., address) and Clinical (e.g., allergies) updates.
- **REQ-PROF-03:** The system shall require user confirmation (Patient or Doctor) before permanently overwriting existing Administrative database fields.
- **REQ-PROF-04:** The system shall maintain an immutable audit log of all profile changes, clearly indicating whether the change was made manually or extracted by the AI.
- **REQ-PROF-05:** The system shall gracefully ignore reports where no new profile data is detected, minimizing unnecessary database write requests.

Feature 16

Feature 16: Patient FAQ & Portal Chatbot

1. Feature Description

The system will provide an embedded Conversational AI Chatbot within the Patient Web Portal. The chatbot acts as a digital concierge, utilizing a RAG (Retrieval-Augmented Generation) pipeline connected strictly to the clinic's public documentation (FAQs, policies, opening hours, portal navigation guides). It is explicitly engineered with strict safety filters to refuse any clinical or diagnostic queries, redirecting patients to book an appointment or contact emergency services if they ask medical questions.

2. Actors Involved

- Primary Actor: Patient
- Secondary System Actor: AI Chatbot Module (LLM + FAQ Vector Database)

3. Use Case Flow: UC-PAT-06 (Query Administrative Chatbot)

- **Precondition:** The Patient is logged into the Web Portal or visiting the clinic's public landing page.
- **Main Success Scenario (Administrative Query):**
 - The Patient opens the chatbot widget and types a question (e.g., *"Where can I park my car?"* or *"Do I need to fast before my blood test tomorrow?"*).
 - The system sends the text to the AI module.
 - The AI first runs the text through a "Medical Intent Filter" to ensure the question is administrative, not diagnostic.
 - The AI queries the Vector Database containing the clinic's official FAQ documents.
 - The AI formulates a friendly response based *only* on the retrieved documents (e.g., *"We have a free parking garage behind the building on Goethestrasse."*).
 - The AI displays the answer to the Patient in the chat window.
- **Alternate Flow (Medical / Diagnostic Query Detected):**
 - The Patient asks a medical question: *"My stomach hurts and I feel dizzy, what should I take?"*

- The Medical Intent Filter flags the input as a clinical query.
- The system intercepts the request *before* searching the database.
- The chatbot outputs a hardcoded safety response: *"I am an administrative assistant and cannot provide medical advice. If you are experiencing a medical emergency, please dial 112 immediately. Otherwise, would you like me to help you book an appointment with a doctor?"*

4. Formal Functional Requirements

- **REQ-CHAT-01:** The system shall provide a conversational text interface for patients to ask administrative and navigational questions.
- **REQ-CHAT-02:** The system's RAG architecture shall be strictly isolated to read only from approved clinic administrative documents, with no access to patient medical records or external medical databases.
- **REQ-CHAT-03:** The system shall implement a semantic safety filter to detect and block queries seeking medical diagnosis, treatment, or triage.
- **REQ-CHAT-04:** The system shall provide hardcoded, legally compliant fallback responses directing patients to emergency services or booking channels when medical intent is detected.
- **REQ-CHAT-05:** The chatbot UI shall display a permanent disclaimer stating it is an AI assistant for administrative purposes only.

Feature 17

Feature 17: Doctor On-Demand Q&A Chatbot

1. Feature Description

The system provides a conversational RAG (Retrieval-Augmented Generation) interface within the Doctor's consultation dashboard. This chatbot allows the Doctor to query the specific patient's entire historical medical record using natural language. The AI acts as a research assistant, retrieving specific data points, summarizing past treatments, and tracking symptom progression over time. To ensure clinical safety, the AI must strictly cite the original source documents for every answer it provides.

2. Actors Involved

- Primary Actor: Doctor
- Secondary System Actor: AI Module (Conversational RAG / LLM)

3. Use Case Flow: UC-DOC-07 (Deep-Dive Query via Chatbot)

- **Precondition:** The Doctor is authenticated, actively viewing a specific Patient's file, and the Patient has historical records in the database.
- **Main Success Scenario:**
 - The Doctor opens the "Copilot Chat" side-panel.
 - The Doctor types a specific query: *"What was the exact date of the patient's last blood test and what were the cholesterol levels?"*
 - The system scopes the search strictly to the current **Patient_ID**.
 - The RAG module queries the vector database, retrieving the relevant chunks of text from the patient's past lab reports.
 - The LLM synthesizes the retrieved chunks into a direct answer.
 - The system displays the answer in the chat UI: *"The last blood test was on October 12, 2025. Total cholesterol was 210 mg/dL."*
 - **Crucial Step:** The system appends a clickable citation below the answer: *[Source: Lab Report - Oct 12, 2025]*.
 - The Doctor clicks the citation to quickly verify the original document if needed.
- **Alternate Flow (Information Not Found):**

- The Doctor asks about a surgery that happened at a different hospital, which is not in Clinic database.
- The RAG module retrieves nothing relevant.
- The LLM is strictly prompted not to guess. It outputs: *"I could not find any information regarding this query in the patient's historical records."*
- **Alternate Flow (Contextual Follow-up):**
 - The Doctor asks a follow-up question: *"Did they prescribe statins after that?"*
 - The system utilizes chat memory to understand "that" refers to the October blood test, and retrieves the corresponding consultation notes to answer accurately.

4. Formal Functional Requirements

- **REQ-DQ-01:** The system shall provide a conversational chat interface capable of maintaining context (memory) throughout a single session.
- **REQ-DQ-02:** The RAG retrieval mechanism shall be strictly sandboxed to the active **Patient_ID** to entirely prevent cross-patient data leakage.
- **REQ-DQ-03:** The LLM shall be configured with a strict system prompt (and low temperature setting) to prioritize factual extraction over creative generation, minimizing hallucination risks.
- **REQ-DQ-04:** The system shall mandate that every AI-generated assertion includes a traceable citation linking back to the original source document in the database.
- **REQ-DQ-05:** The system shall explicitly state when information cannot be found within the provided context, rather than attempting to infer an answer.

Feature 18

Feature 18: AI-Assisted Staff Scheduling & Conflict Resolution

1. Feature Description

The system utilizes an AI Optimization Module to assist the Clinic Manager in handling staff rosters and sudden schedule disruptions. When a Doctor becomes unexpectedly unavailable (e.g., sick leave), the AI analyzes the clinic's active calendar, evaluates the availability and workload of other Doctors, and automatically proposes an optimized recovery plan. This plan suggests which appointments can be safely reassigned to other colleagues and which must be cancelled, minimizing patient disruption and protecting clinic revenue.

2. Actors Involved

- Primary Actor: Receptionist (or Clinic Manager)
- Secondary System Actor: AI Scheduling Module (Constraint Optimization Logic)

3. Use Case Flow: UC-REC-05 (AI-Assisted Sick Leave Resolution)

- **Precondition:** The Receptionist is authenticated, and a Doctor (Dr. Smith) calls in sick for the day. Dr. Smith has 15 scheduled appointments.
- **Main Success Scenario:**
 - The Receptionist navigates to the "Staff Scheduling" dashboard and marks Dr. Smith as **Unavailable - Sick** for the day.
 - The system detects a critical calendar conflict (15 orphaned appointments).
 - The system automatically triggers the AI Scheduling Module.
 - The AI analyzes the schedules of the other doctors on duty (e.g., Dr. Müller and Dr. Weber).
 - The AI discovers Dr. Müller has an administrative block that can be moved, and Dr. Weber has two empty time slots.
 - The AI generates a "Proposed Recovery Plan" and displays it to the Receptionist:
 - *Reassign 10 patients to Dr. Müller.*
 - *Reassign 2 patients to Dr. Weber.*

- *Cancel and reschedule the remaining 3 patients.*
- The Receptionist reviews the AI's logic, makes any manual adjustments, and clicks "Execute Plan."
- The system instantly updates the database: transferring the 12 appointments to the new doctors and sending automated SMS updates to all 15 patients explaining the changes.
- **Alternate Flow (No Available Replacements):**
 - The AI analyzes the clinic and sees all other doctors are operating at 100% capacity.
 - The AI outputs: *"No available coverage. All 15 appointments must be cancelled."*
 - The Receptionist clicks "Batch Cancel," and the system frees the time slots and notifies the patients.

4. Formal Functional Requirements

- **REQ-SCHED-01:** The system shall maintain a centralized matrix of all Doctor availability, maximum workload capacities, and scheduled appointments.
- **REQ-SCHED-02:** The system shall utilize an AI optimization algorithm to dynamically propose appointment reassignments when schedule conflicts occur.
- **REQ-SCHED-03:** The AI module shall prioritize patient retention (reassigning to colleagues) over appointment cancellation whenever mathematically possible.
- **REQ-SCHED-04:** The system must enforce a "Human-in-the-Loop" requirement, mandating that the Clinic Manager explicitly approve the AI's proposed schedule changes before they are committed to the database.
- **REQ-SCHED-05:** Upon approval of a schedule change, the system shall automatically trigger notification workflows to inform affected patients of new providers or cancellations.

WON'T HAVE

Feature 19

Feature 19: Internal Staff Onboarding AI Chatbot

1. Feature Description

The system will eventually provide an internal conversational AI tool specifically designed to assist newly hired administrative staff. Utilizing a **RAG** architecture connected strictly to the clinic's internal Standard Operating Procedures (SOPs) and software manuals, the chatbot will answer operational questions (e.g., "How do I process a refund?" or "What is the policy for walk-in emergencies?"), reducing the training burden on the Clinic Manager.

2. Actors Involved

- Primary Actor: Receptionist (Trainee / New Hire)
- Secondary System Actor: AI Module (Internal **RAG** / **LLM**)

3. Use Case Flow: UC-STF-01 (Query Internal SOPs)

- **Precondition:** A newly hired Receptionist is authenticated and encounters an unfamiliar administrative task.
- **Main Success Scenario:**
 - The Receptionist opens the "Staff Help" chatbot widget.
 - The Receptionist types a question: *"How do I register a patient who has no physical address?"*
 - The AI runs a vector search exclusively against the clinic's internal training manuals and HR policies.
 - The LLM extracts the relevant policy and formulates an answer.
 - The AI displays the answer: *"According to clinic's SOP 4.2, you must enter the clinic's address as a placeholder and tag the profile for Manager Review."*
 - The AI provides a clickable link to "SOP 4.2".
 - The Receptionist successfully completes the task without having to interrupt a senior staff member.
- **Alternate Flow (No Policy Found):**
 - The Receptionist asks a highly specific or undocumented question.
 - The AI searches the vector database and finds no relevant SOP.

- The AI outputs: *"I cannot find a documented procedure for this specific scenario. Please consult the Clinic Manager for guidance."*

4. Formal Functional Requirements

- **REQ-TRN-01:** The system shall provide a text-based conversational interface dedicated to internal staff support.
- **REQ-TRN-02:** The RAG database for this module shall be strictly isolated from all patient health data, containing only administrative, HR, and software documentation.
- **REQ-TRN-03:** The AI shall explicitly cite the internal document or manual it used to generate its response.
- **REQ-TRN-04:** The system shall actively log unanswered queries (Alternate Flow) to notify management of gaps in their internal training documentation.
- **REQ-TRN-05: [ROADMAP DEFERRAL]** Implementation of this module is explicitly deferred to Version 2.0, allowing V1.0 resources to focus on patient-facing and clinical AI features.

SOP: Standard Operating Procedure